

Human Performance Modeling - 2

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Elevating Input Technology in the Era of AI

Challenge :

Real user input data are **sparse**.

Reasons:

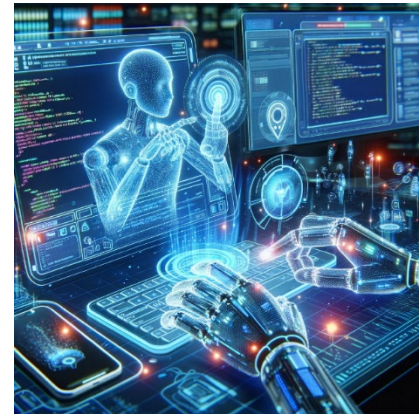
- Privacy Concerns
- Sparse Input Events



Solution:

Generative AI Models for User Input

- Simulate user input and interaction events



Can ChatGPT 4
generate typing data?



You

Generate a picture of the gesture trace for entering word "nice" with gesture typing on a Qwerty soft keyboard.



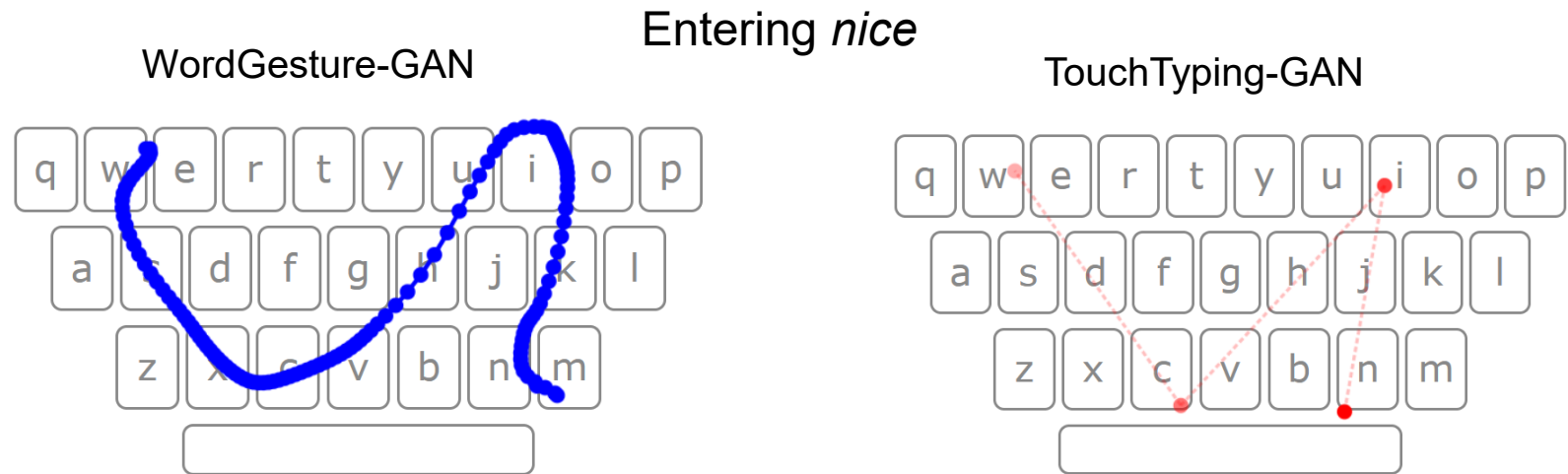
ChatGPT



Here's the illustration showing the gesture trace for typing the word "nice" on a QWERTY soft keyboard displayed on a smartphone screen. The tracing path connects the letters N, I, C, and E, demonstrating the gesture typing method.

WordGesture-GAN and TouchTyping-GAN

- Generating Typing Input using Generative Adversarial Networks

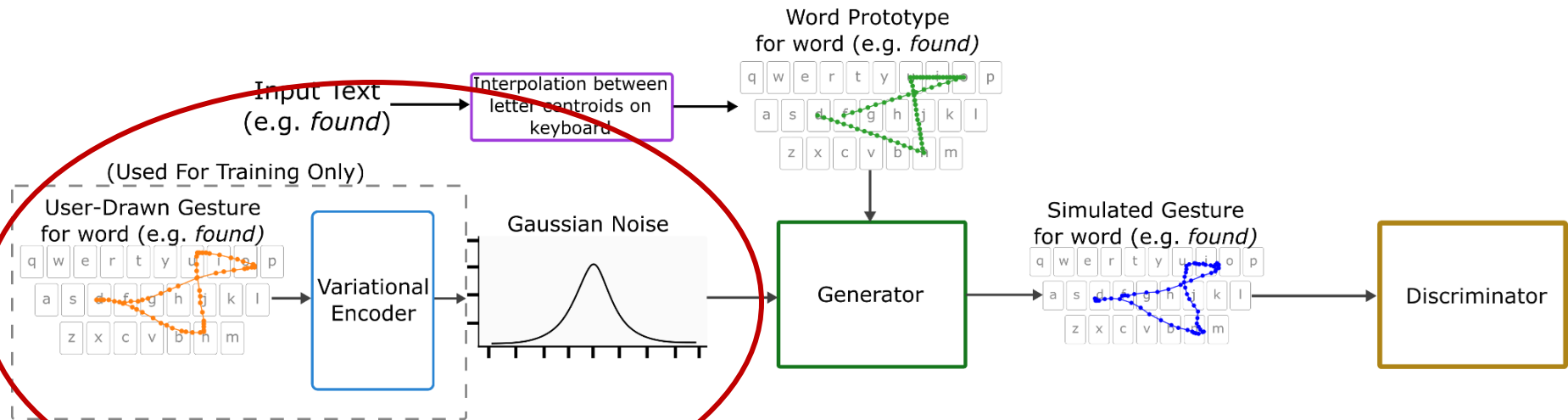


J. Chu, D. An, Y. Ma, W. Cui, S. Zhai, X. Gu, **X. Bi** "WordGesture-GAN: Modeling Word-Gesture Movement with Generative Adversarial Network ". *CHI 2023*. Article No.: 287 Pages 1–15.

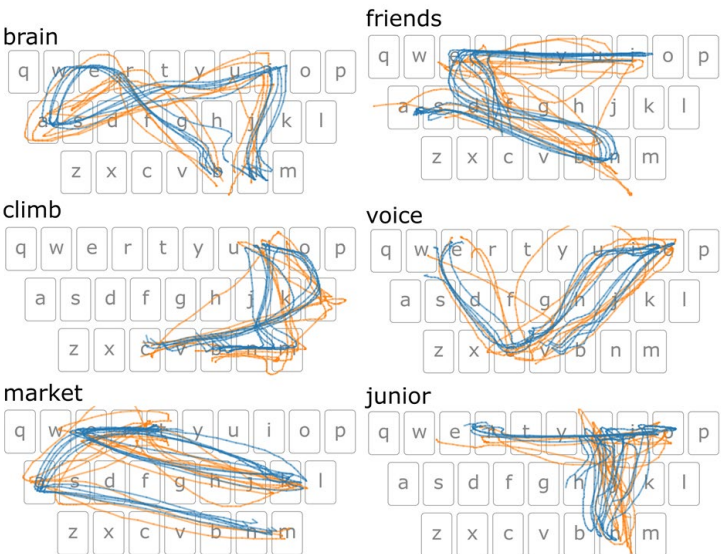
J. Chu, Y. Ma, S. Zhai, X. Gu, **X. Bi** "TouchTyping-GAN: Modeling Touch Typing with Generative Adversarial Network ". *UIST 2023*. Article 29. Pages 1 - 13

Architecture of WordGesture-GAN

- Using a Variational Encoder to control the latent space.



User-drawn (Orange) vs. Simulated (Blue)



Training with simulated data improves the accuracy of gesture typing decoder.

Training Setup (# of gestures)	Error Rate After Training
200 user-drawn	32.8%
10000 simulated	28.6%
10000 user-drawn	27.8%

Our GAN models are adopted by

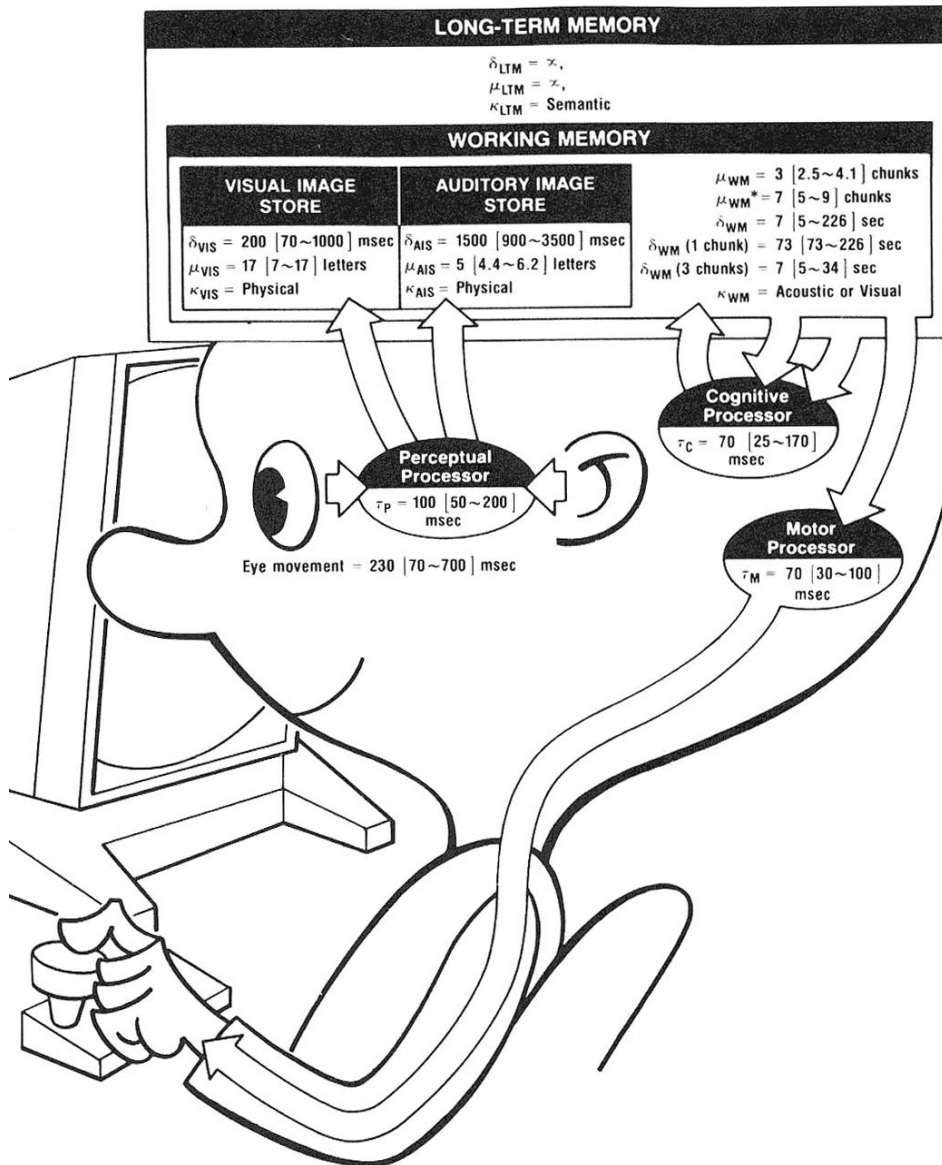


Google's Gboard



Grammarly Keyboard

Human Information Processor



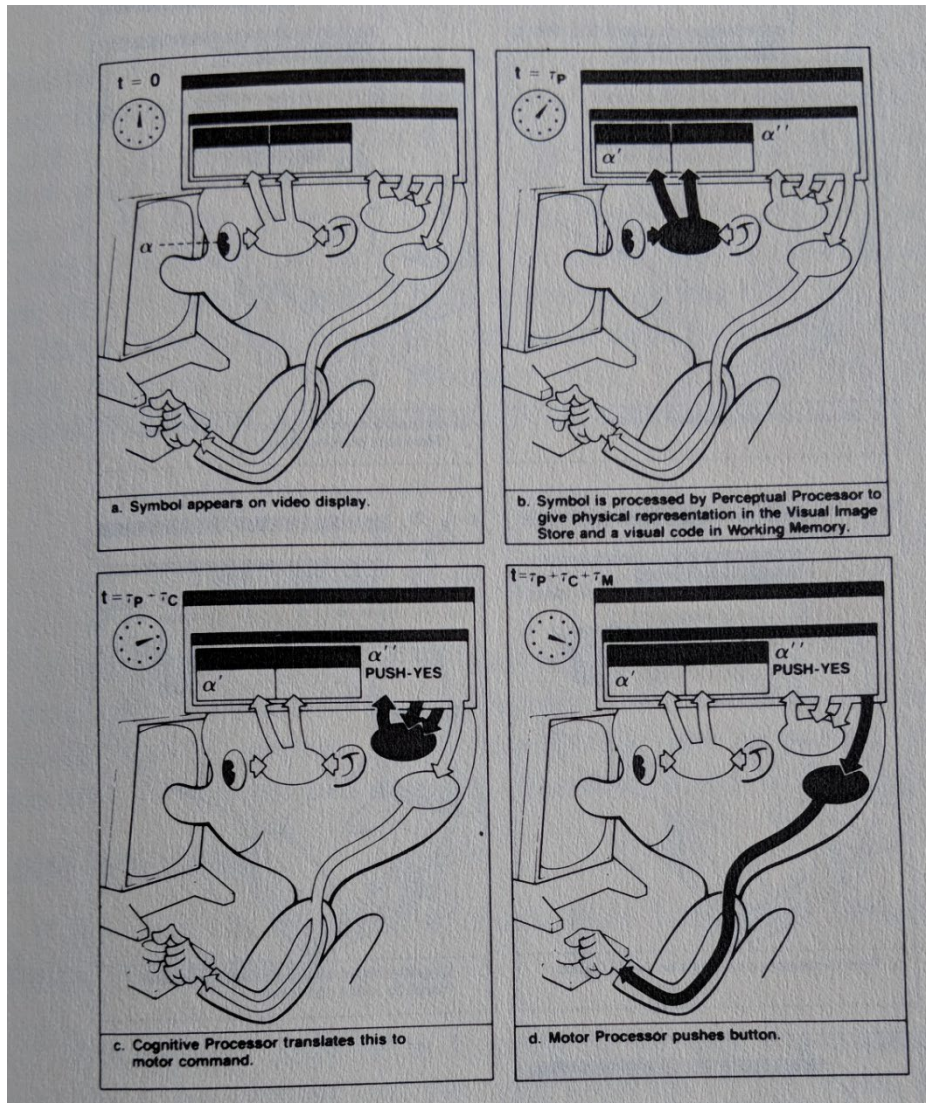
- Each subsystem has its own memories and processors.
- Memory
 - μ , storage capacity in items
 - δ , decay time of an item
- Processor
 - τ , cycle time
- Three subsystems can work in parallel.

Simple Decisions

Questions

A user sits before a computer display terminal. Whenever any symbol appears, he is to press the space bar. What is the time between signal and response?

Questions



Reaction Time =

$$\tau_p + \tau_c + \tau_m = 100 + 70 + 70 = 240 \text{ ms}$$

Today's Questions

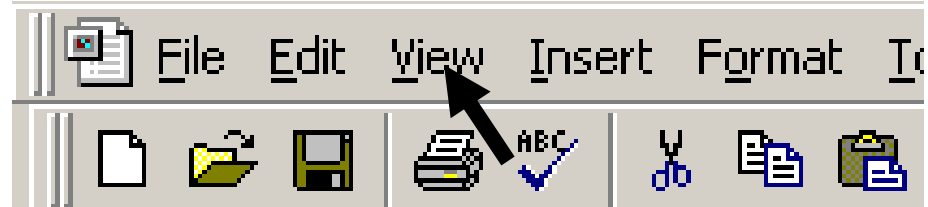
You have a palette of tools in a graphics application that consists of a matrix of 20x16-pixel icons laid out as a 2x8 array that lies along the left-hand edge of the screen. Without moving the array from the left-hand side of the screen or changing the size of the icons, what steps can you take to decrease the time necessary to access the average tool?



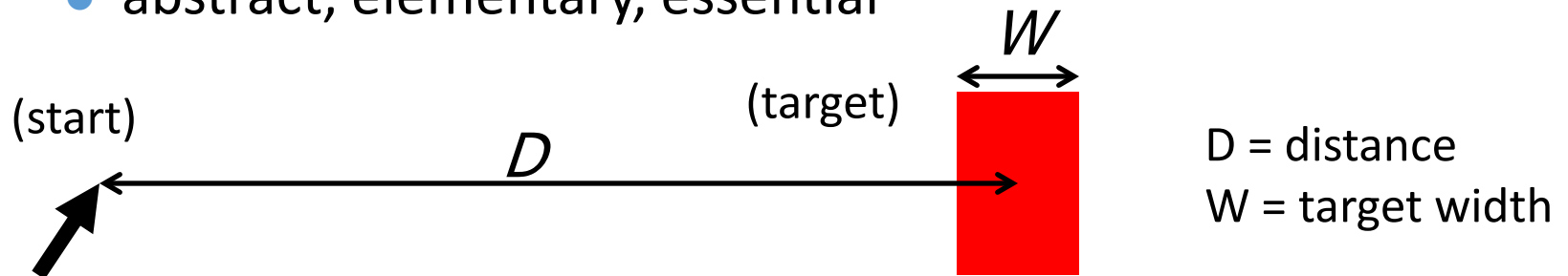
Motor Skills

Pointing Tasks

- Real task: interacting with GUI's
 - Pointing is fundamental



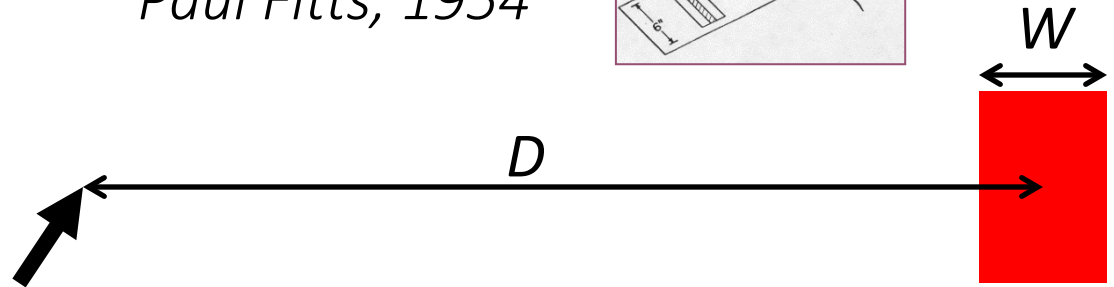
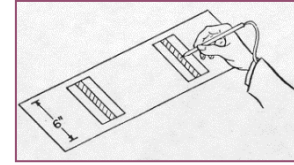
- Experimental task: target acquisition
 - abstract, elementary, essential



D is also referred to as A (amplitude) in literature.

Fitts' Law

Paul Fitts, 1954



$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Movement Time

Index of Difficulty ($ID [bits]$)

The original Fitts' law proposed by Paul Fitts in 1954:

$$MT = a + b \log_2 \left(\frac{2D}{W} \right)$$

Both are valid:

$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

$$MT = a + b \log_2 \left(\frac{2D}{W} \right)$$

$$\text{Index of Difficulty (ID)} = \log_2 \left(\frac{D}{W} + 1 \right), \text{ or } \log_2 \left(\frac{2D}{W} \right)$$

Iterative-Correction Model

It assumes that movement consists of n sub-movement:

$$MT = nt.$$

Each sub-movement moves a constant fraction of the remaining distance to the center of the target

$\frac{X_i}{X_{i-1}} = K,$ ← The remaining distance after i th sub-movement

A series movement are made until it reaches the target:

$$X_n = KX_{n-1} = K^2X_{n-2} = \dots = K^n A = \frac{1}{2}W.$$

Iterative-Correction Model

Solving for n , we have

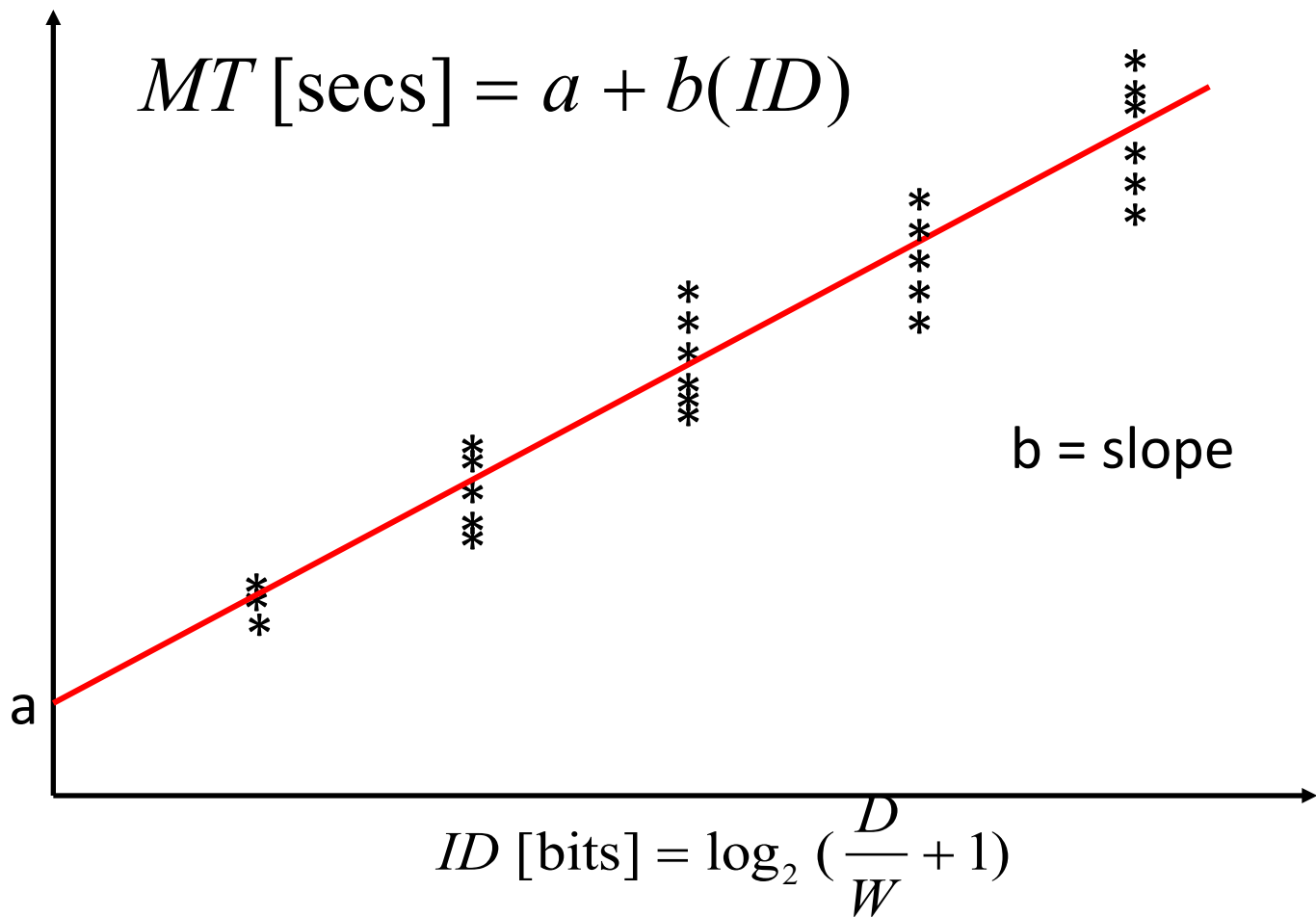
$$n = -\frac{1}{\log_2 K} \cdot \log_2\left(\frac{2A}{W}\right) = -\frac{1}{\log_2 K} \cdot ID.$$

Therefore:

$$MT = nt = b \cdot \log_2\left(\frac{2A}{W}\right) = b \cdot ID,$$

Assuming there is initial movement time:

$$MT = a + nt = a + b \cdot \log_2\left(\frac{2A}{W}\right) = a + b \cdot ID,$$

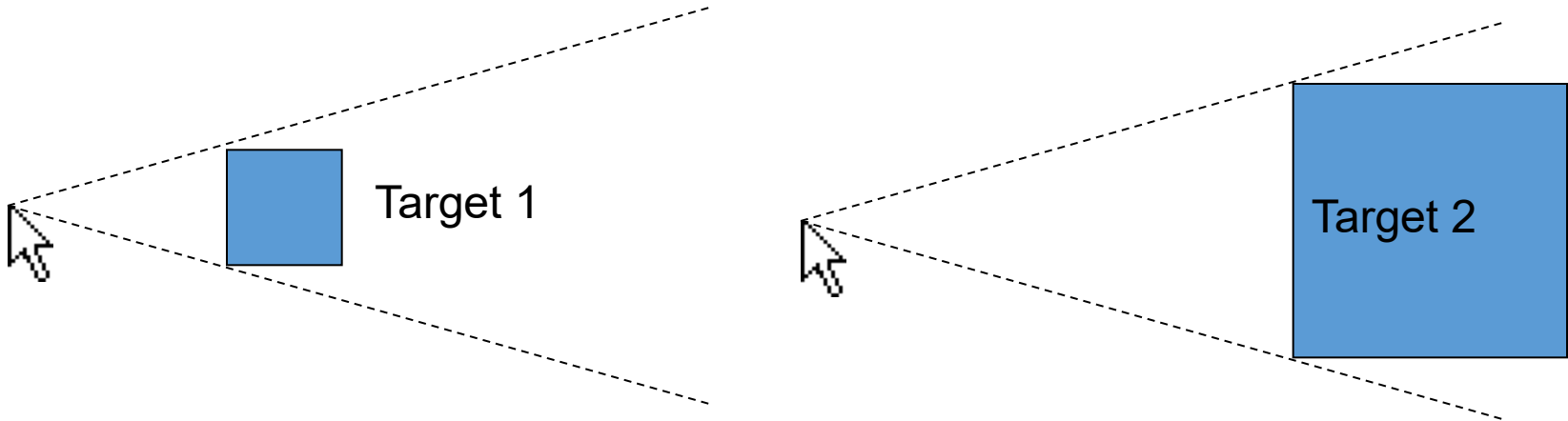


$$MT = a + b \log_2 \left(\frac{D}{W} + 1 \right)$$

Only Relative Precision (D / W) Matters

Fitts' Law

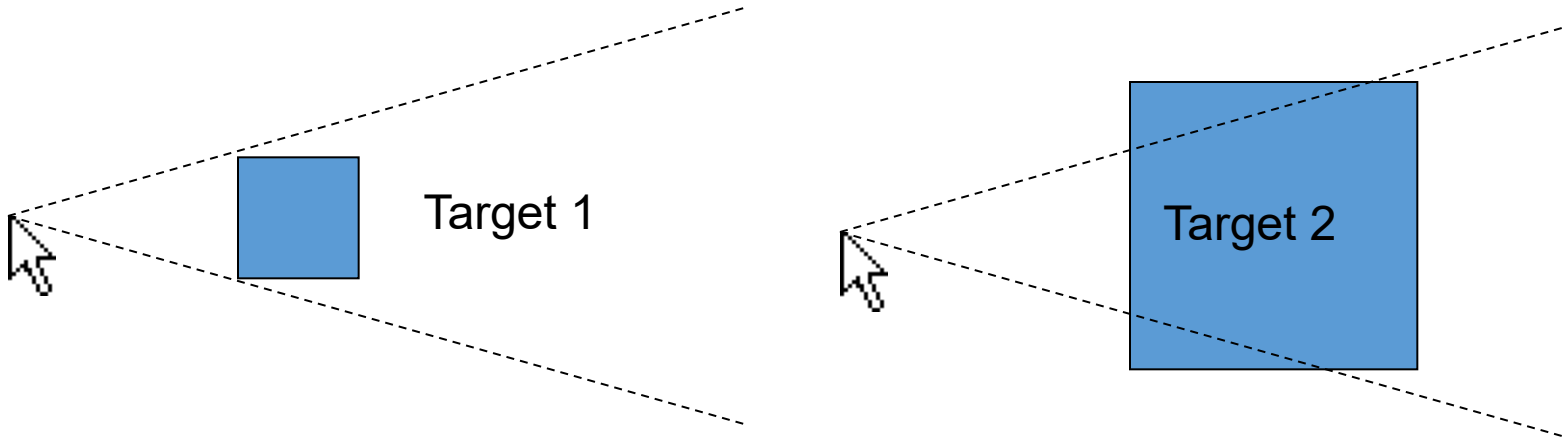
$$\text{Time} = a + b \log_2(D/W+1)$$



Same ID $\rightarrow$ Same Difficulty

Fitts' Law

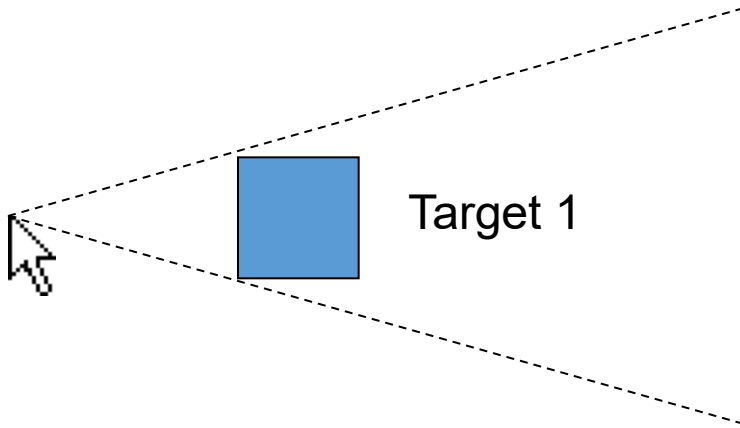
$$\text{Time} = a + b \log_2(D/W + 1)$$



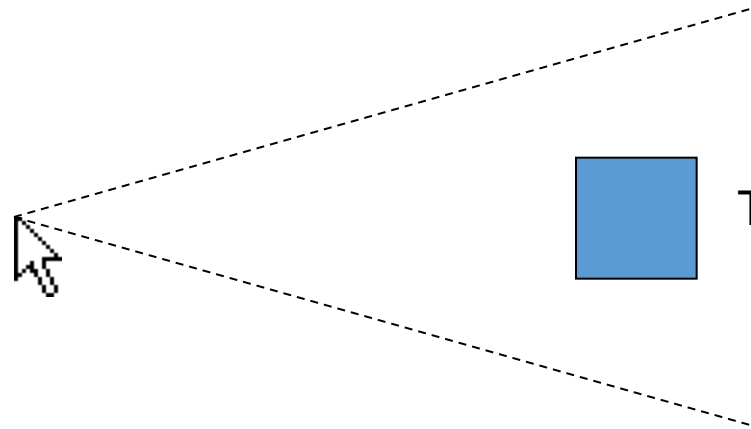
Smaller ID $\rightarrow$ Easier

Fitts' Law

$$\text{Time} = a + b \log_2(D/W + 1)$$



Target 1



Target 2

Larger ID $\rightarrow$ Harder